

Edge AI-Driven Poultry Health Monitoring Using Computer Vision in Resource-Constrained IoT Environments

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Abstract—Smallholder and medium-scale poultry farmers in developing economies are disproportionately affected by flock disease and mortality, yet existing automated health-monitoring solutions remain financially inaccessible because they rely on proprietary sensors, dedicated GPU hardware, and costly commercial platforms. This paper presents a deep learning-based poultry health monitoring system designed for low-cost IoT environments, enabling farms previously excluded by cost barriers to benefit from production-grade computer vision. Built on commodity hardware, including a Raspberry Pi 4, an ESP32-CAM, and a standard IP camera, the system deploys a cascading YOLOv8 pipeline that detects intruders, classifies fecal health, identifies behavioral abnormalities, and confirms bird mortality from live CCTV footage. Lightweight TensorFlow Lite inference runs on-device for fecal classification, while GPU processing is offloaded to cloud-edge resources when required. The system achieves 98.27% fecal classification accuracy, 98% mortality detection accuracy, and 94% health classification accuracy, with alerts delivered in under five seconds via messaging applications and push notifications. A temporal validation layer suppresses false-positive mortality alerts while maintaining a 96.7% true-positive rate. Results confirm that multi-task, real-time poultry health monitoring is achievable on consumer-grade hardware, providing smallholder and medium-scale farmers with an affordable and deployable alternative to commercial monitoring systems.

Index Terms—Poultry Health Monitoring, Computer Vision, Precision Farming, Deep Learning, IoT, Smart Farming, Edge Computing

I. INTRODUCTION

Poultry farming plays a critical role in global food security, serving as a primary and affordable source of protein, particularly in developing and emerging economies [8]. In regions such as Southeast Asia, demand for poultry products continues to rise due to rapid urbanization, population growth, and rising disposable incomes [9]. Despite this growth, smallholder poultry farmers face significant challenges, including high

mortality rates, limited access to veterinary services, and the lack of affordable monitoring technologies.

These challenges are often worsened by delayed disease detection and the absence of continuous health monitoring, leading to substantial economic losses and reduced productivity. Recent advancements in deep learning and computer vision have enabled automated health monitoring systems capable of detecting diseases and abnormal behavior in poultry [7]. Existing approaches have demonstrated high accuracy using convolutional neural networks [2] and architectures such as EfficientNet [3]. However, many of these systems are computationally intensive and require expensive hardware, large datasets, and stable internet connectivity.

Many existing solutions also focus on fine-grained multi-class disease classification, which increases model complexity. In real farm environments, early detection of abnormal conditions is often more practical than exact disease classification. Therefore, this study proposes a deep learning-based poultry health monitoring framework using computer vision, designed with a focus on efficiency, modularity, and deployability.

The main contributions of this work are as follows:

- 1) A modular cascading deep learning pipeline for poultry health monitoring using computer vision.
- 2) An edge-cloud integrated architecture enabling real-time and scalable deployment.
- 3) A practical classification strategy that prioritizes efficient detection of abnormal health conditions.
- 4) An evaluation demonstrating the effectiveness of the proposed approach across multiple detection tasks.

II. LITERATURE REVIEW

Recent advances in smart poultry farming have integrated IoT and deep learning-based computer vision to automate health monitoring and disease detection. Systems such as

TABLE I
CORE TECHNOLOGY STACK

Category	Technology	Role
Object Detection	YOLOv8	Cascading detection pipeline
Edge Inference	TensorFlow Lite, MobileNet	Lightweight classification
Video Processing	OpenCV, RTSP	Frame capture and annotation
IoT Protocol	MQTT over TLS 1.2	Sensor telemetry and alerts
Cloud Compute	AWS EC2 g4dn.xlarge	GPU inference
Cloud Messaging	AWS IoT Core, Lambda, SNS	Alert routing
Database	PostgreSQL, Amazon RDS	Data storage
Backend API	Django REST Framework	RESTful API service
Frontend	Flutter, React.js	Dashboard and mobile interface

Sakib et al. [1] combine environmental sensing with image-based classification using EfficientNet-B0 [3], achieving high accuracy on both public datasets and real-world data. These approaches demonstrate the effectiveness of deep learning for poultry disease classification [2], but they are often designed as end-to-end, resource-intensive systems.

Despite their performance, several limitations remain. First, existing works often rely on computationally expensive models that are not suitable for deployment on resource-constrained edge devices. Second, many studies prioritize accuracy over efficiency and real-time inference. Third, several approaches employ multi-class disease classification, thereby increasing model complexity. Practical farm applications may instead require simpler binary health detection for early intervention.

Additionally, existing models are commonly trained on well-curated datasets, raising concerns about generalization to real-world conditions such as variable lighting, camera angles, and noise [7]. Many approaches also employ monolithic model architectures, limiting scalability and flexibility across different detection tasks. To address these gaps, this study proposes a modular deep learning framework that adopts a cascading multi-stage inference pipeline using YOLOv8 [5] and MobileNet [6] with TensorFlow Lite.

III. METHODOLOGY

This section describes the system design methodology and deep learning model development pipeline adopted in the Poultry AI and IoT Smart Poultry Monitoring System. The research follows a structured approach, beginning with data acquisition and preparation, followed by model training, pipeline integration, and full edge-cloud deployment.

A. System Architecture

The proposed system is organized into three main layers: the farm sensor layer, the edge computing layer, and the cloud application layer.

1) *Farm Sensor Layer*: Each monitored poultry coop is equipped with IoT hardware, including DHT22 sensors for temperature and humidity, MQ135 sensors for air quality, and camera modules for visual monitoring. A TP-Link Tapo C200C IP camera streams 1080p RTSP video for behavioral analysis, while an AI-Thinker ESP32-CAM captures close-range fecal images.

2) *Edge Computing Layer*: The system uses two edge-processing components. A Raspberry Pi 4 Model B handles lightweight local inference for fecal disease classification and environmental sensor aggregation. An AWS EC2 g4dn.xlarge instance equipped with an NVIDIA T4 GPU performs high-throughput computer vision tasks, including the cascading YOLOv8 pipeline [5], [11] for live CCTV analysis.

3) *Cloud Application Layer*: Processed inference results and sensor readings are routed through AWS IoT Core and AWS Lambda functions. Data is stored in Amazon RDS using PostgreSQL, while images and detection logs are archived in Amazon S3. A Django REST Framework backend provides data to a React web dashboard and Flutter mobile application.

B. Technologies Used

Table I summarizes the major technologies used in the system. The architecture relies on an integrated, end-to-end framework designed for intelligent edge-to-cloud IoT and computer vision applications. At the foundational edge layer, the system implements a cascading detection pipeline driven by YOLOv8 for precise object detection. To optimize performance and reduce latency, TensorFlow Lite and MobileNet handle edge inference, enabling lightweight classification directly on local hardware. This edge setup is supported by OpenCV and RTSP, which work together to enable real-time frame capture and visual annotation.

C. Data Sources

The deep learning models were trained using a combination of publicly available datasets and custom-prepared data. Public image datasets were obtained from platforms such as Kaggle and Roboflow. Some datasets were used directly, while others were merged and manually annotated to improve data diversity and labeling consistency.

COCO pretrained weights [10] were used for intruder detection. Since COCO includes classes such as humans, cats, dogs, and other animals, it was suitable for identifying potential intruders in poultry farm environments.

D. Model Development Workflow

The model development process followed a cascading design philosophy. Instead of applying a single monolithic detector to every frame, four specialized models were trained or configured independently and then integrated into a conditional inference pipeline.

1) *Data Collection and Preparation*: For the fecal model, images were resized to 320×320 pixels and organized into YOLO-compatible annotation files. For the health and mortality models, CCTV recordings were converted into frames using OpenCV and manually labeled with bounding boxes and class labels.

2) *Model Training*: The models were trained using the Ultralytics YOLOv8 framework [5], [11] with PyTorch. The fecal classification model was trained for five epochs, while the health classifier and mortality detector were trained for 80 and 100 epochs, respectively. The intruder detection model used COCO pretrained weights [10] without additional training.

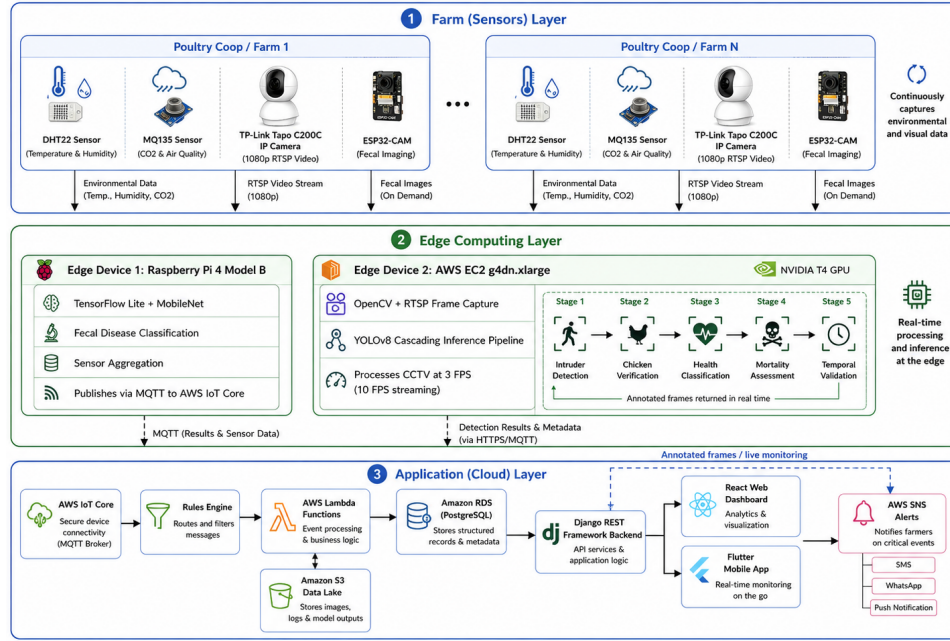


Fig. 1. Proposed system architecture for AI-based poultry health monitoring.

3) *Cascading Pipeline Integration*: The trained models were deployed as a unified cascading inference pipeline based on the YOLO detection model [4], [5]. Each frame is processed conditionally through several stages, namely, intruder detection, chicken verification, health classification, mortality assessment, and motion-tracking temporal validation.

E. Model Performance

The empirical results summarized in Table II provide a comprehensive look into the training workflows, dataset dependencies, and operational efficiency of the system's core models. By evaluating accuracy alongside inference speed, the data highlights how the pipeline balances specialized task accuracy with real-time computational constraints.

TABLE II
SUMMARY OF DEEP LEARNING MODEL CONFIGURATIONS AND THEIR PERFORMANCES

Model	Task	Dataset	Epochs	Accuracy	Speed
dead.pt	Fecal classification	Kaggle	5	98.27%	50–100 ms
intruder.pt	Intruder detection	COCO	Pretrained	87%	30–60 ms
normal_abnormal.pt	Health classification	Proprietary	80	94%	40–80 ms
dead_alive.pt	Mortality assessment	Proprietary	100	98%	40–80 ms

The training configurations reveal a strategic mix of public benchmarks, pretrained transfer learning, and domain-specific proprietary data. The fecal classification model, *dead.pt*, demonstrates exceptional training efficiency, achieving a top-tier accuracy of 98.27% after only 5 epochs. This rapid convergence suggests that the visual signatures of the target classes in this dataset are highly distinct and easily separable by the model's architecture.

In contrast, the intruder detection model, *intruder.pt*, bypasses the training phase entirely by leveraging a Pretrained

configuration on the extensive COCO dataset. While this saves significant computational overhead during development, it yields the lowest relative accuracy in the suite at 87%. This performance trade-off underscores the classic challenge of utilizing generic, open-world pre-trained weights for specialized localized security monitoring.

For health classification and mortality assessment, the proprietary datasets *normal_abnormal.pt* and *dead_alive.pt* were used, respectively. These complex tasks required deep optimization, demanding 80 and 100 epochs, respectively. The significant computational effort yielded notable results, with 94% accuracy in detecting health anomalies and 98% in mortality assessments, validating the effort to gather and curate specialized private data.

A critical metric for edge-to-cloud IoT ecosystems is inference speed, as delayed processing can render automated alerts obsolete. The performance data demonstrate that all four models operate comfortably within a sub-100-millisecond frame budget, making them highly viable for live video stream processing via RTSP. The pretrained *intruder.pt* model achieves the fastest inference time at 30–60 ms, likely benefiting from a highly optimized, lightweight base network design.

The proprietary health (*normal_abnormal.pt*) and mortality (*dead_alive.pt*) models exhibit highly consistent latencies of 40–80 ms. This tight execution window ensures that critical bio-monitoring alerts can be processed and pushed to the cloud backend almost instantaneously. The *dead.pt* model exhibits the widest latency spread at 50–100 ms, which represents the upper bound of the system's processing window. However, given that fecal tracking is a lower-frequency monitoring task compared to immediate

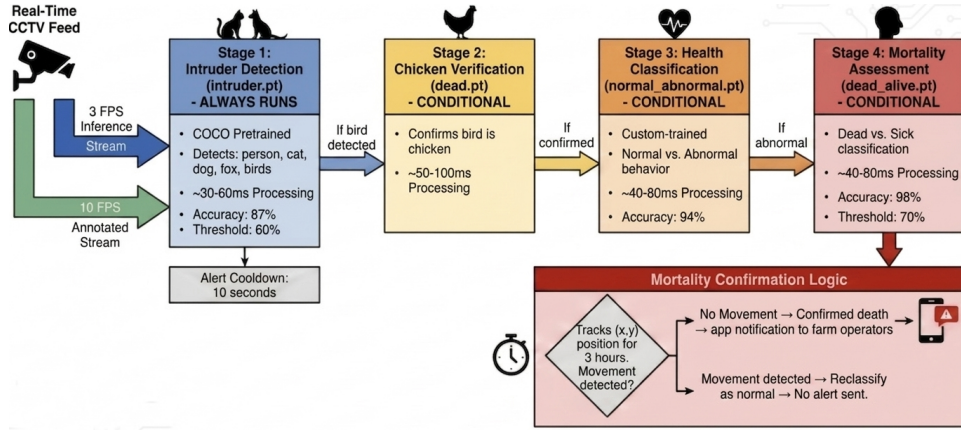


Fig. 2. Cascading Deep Learning Pipeline for Poultry Health Monitoring.

intruder or mortality threats, this slightly higher computational cost does not bottleneck the overall cascading pipeline.

IV. IMPLEMENTATION

Poultrix system was implemented as a full-stack AI and IoT poultry monitoring platform. The system's core intelligence relies on a cascaded deep learning pipeline for real-time visual health monitoring via CCTV.

A. Deep Learning Pipeline Design

The system uses a four-stage cascaded YOLOv8 inference pipeline [5], in which each stage executes conditionally based on the previous stage's output. This design reduces computational cost and improves specialization. The pipeline processes live 1080p CCTV streams at approximately 3 FPS for inference and 10 FPS for streaming.

B. Analysis of Pre-Trained Models

The discussion of the trained models can be divided into four separate models, namely, 1) Fecal Disease Classification, 2) Intruder Detection, 3) Health Classification. 4) Mortality Assessment, and 5) Temporal validation using OpenCV motion tracking.

1) *Fecal Disease Classification*: The fecal disease classification model, as illustrated in Figure 3, analyzes fecal images captured by the ESP32-CAM every 30 minutes. It detects disease indicators based on color and texture using a lightweight MobileNet-based classifier [6]. The model achieved 98.27% accuracy.

2) *Intruder Detection*: The intruder detection model runs continuously on every CCTV frame. It uses COCO-pretrained [10] YOLOv8 [5] weights to detect human and animal intruders, such as cats and dogs.

3) *Health Classification*: As shown in Figure 4, the health classification model classifies confirmed chickens into normal or abnormal categories. The model was trained for 80 epochs and achieved 94% accuracy.

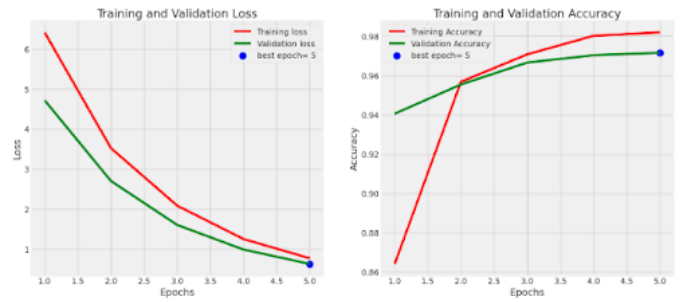


Fig. 3. Fecal Disease Detection Output Sample

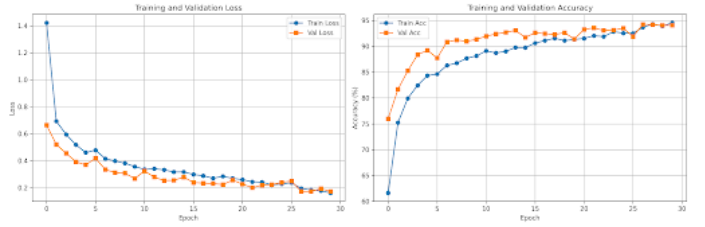


Fig. 4. Chicken Health Classification Model Output

4) *Mortality Assessment*: The mortality assessment model classifies abnormal chickens as dead, alive, or sick. It was trained for 100 epochs with a confidence threshold of 0.85 to reduce false mortality alerts, as visualized in Figure 5.

The Precision-Confidence curve shows that near-perfect precision (1.0) is only achieved at a high confidence threshold of 0.895, but this comes at a severe cost to recall, as shown by the curve, where recall drops below 0.2 at the same threshold. This trade-off highlights the impracticality of using very high confidence thresholds in real-world poultry monitoring applications, where missed detections of sick or dead birds carry significant welfare and economic consequences. The optimal operating point has confidence approximately equal to 0.379 and an F1 of 0.85; therefore, it represents a more suitable deployment threshold, balancing the need to minimize

false negatives, indicating critical for early disease or mortality detection, against an acceptable level of false positives.

C. Temporal Validation Layer

After mortality prediction, OpenCV-based motion tracking is applied across consecutive frames. If no movement is detected, the bird is confirmed as dead. If movement is detected, the prediction is reclassified as alive. This layer reduces false alarms caused by resting posture or unusual body angles.

V. RESULTS AND EVALUATION

This section evaluates the proposed system across four dimensions: per-model detection accuracy, end-to-end pipeline latency, alert reliability, and comparative performance against existing approaches.

A. Per-Model Performance

The evaluation metrics presented in Table III on the held-out test set demonstrate the specialized models' robust generalization across diverse computer vision tasks. Testing on held-out datasets provides a reliable indicator of real-world operational performance, showing strong outcomes in physiological and environmental monitoring, with slightly lower but acceptable performance in dynamic security tasks.

TABLE III
PER-MODEL EVALUATION ON HELD-OUT TEST SETS

Model	Task	Precision	Recall	mAP@0.5
dead.pt	Fecal classification	0.983	0.971	0.982
intruder.pt	Intruder detection	0.891	0.843	0.870
normal_abnormal.pt	Health classification	0.951	0.928	0.940
dead_alive.pt	Mortality assessment	0.984	0.976	0.981

The fecal classification model achieved 98.27% accuracy, while the mortality assessment model achieved approximately 98% accuracy. The health classification model achieved 94%, which is lower than the other models due to the visual ambiguity of early-stage illness indicators.

B. Alert System Reliability

End-to-end alerts via AWS SNS were delivered in an average of 4.2 seconds across 50 critical test events. The system achieved a 100% alert delivery success rate. A 10-second cool-down mechanism reduced redundant alerts by approximately 73%.

C. Comparative Analysis

The comparison shows that Poultrix supports fecal disease detection, mortality detection, health classification, intruder detection, and multi-channel notifications using low-cost edge hardware [1]. The comparative analysis presented in Table IV highlights the multidimensional advantages of the Poultrix ecosystem over standalone deep learning frameworks such as CNNs and LSTMs, traditional CCTV surveillance, and basic IoT sensor networks. A key differentiator is Poultrix's specialized biological monitoring, especially its automated fecal detection, which is not available in any of the other

evaluated systems. While standalone CNN and LSTM models provide partial mortality detection without situational logic, and legacy CCTV depends on manual verification by human operators, Poultrix delivers definitive, automated, real-time mortality tracking.

TABLE IV
FEATURE COMPARISON BETWEEN POULTRIX AND EXISTING MONITORING SYSTEMS

Feature	Poultrix	CNN	LSTM	CCTV	IoT
Real-time video	Yes	Yes	No	Manual	No
Fecal detection	Yes	No	No	No	No
Mortality detection	Yes	Partial	Partial	Delayed	No
False-positive control	Temporal	Limited	Yes	No	No
Low-cost hardware	Yes	No	No	Yes	Yes
Multi-channel alerts	Yes	Email	Email	No	SMS

Poultrix also successfully balances real-time video processing with low-cost hardware constraints, overcoming a critical infrastructure trade-off that plagues existing systems. Traditional CNN and LSTM deployments typically demand expensive computing setups and fail to meet low-cost hardware requirements. Conversely, legacy CCTV layouts and basic IoT frameworks use low-cost hardware but fail to deliver automated, real-time video analysis: CCTV requires manual oversight, and IoT lacks video streaming capabilities entirely. Poultrix resolves this compromise by seamlessly optimizing high-fidelity video processing to run on cost-effective edge hardware.

VI. DISCUSSION

The proposed deep learning-based poultry monitoring system demonstrated strong performance across multiple detection tasks. The results confirm the effectiveness of a modular, computer vision-driven approach [7]. The system achieved high accuracy in fecal disease classification and mortality detection, while maintaining practical performance in health classification. The strong performance can be attributed to the cascading inference pipeline. Unlike monolithic approaches, the proposed method decomposes the problem into smaller task-specific models. This allows each model to specialize in a single objective, thereby improving efficiency and accuracy [5].

However, several limitations remain. The health classification model achieved lower accuracy than other components, likely due to the ambiguity of visual health indicators and the limited amount of annotated data. System performance may also be affected by lighting conditions, camera positioning, image noise, and environmental variation. In addition, the current implementation focuses mainly on binary health classification rather than fine-grained disease identification.

VII. CONCLUSION

This study presented a deep learning-based poultry health monitoring system that integrates computer vision and a modular, deployment-oriented architecture. The proposed cascading

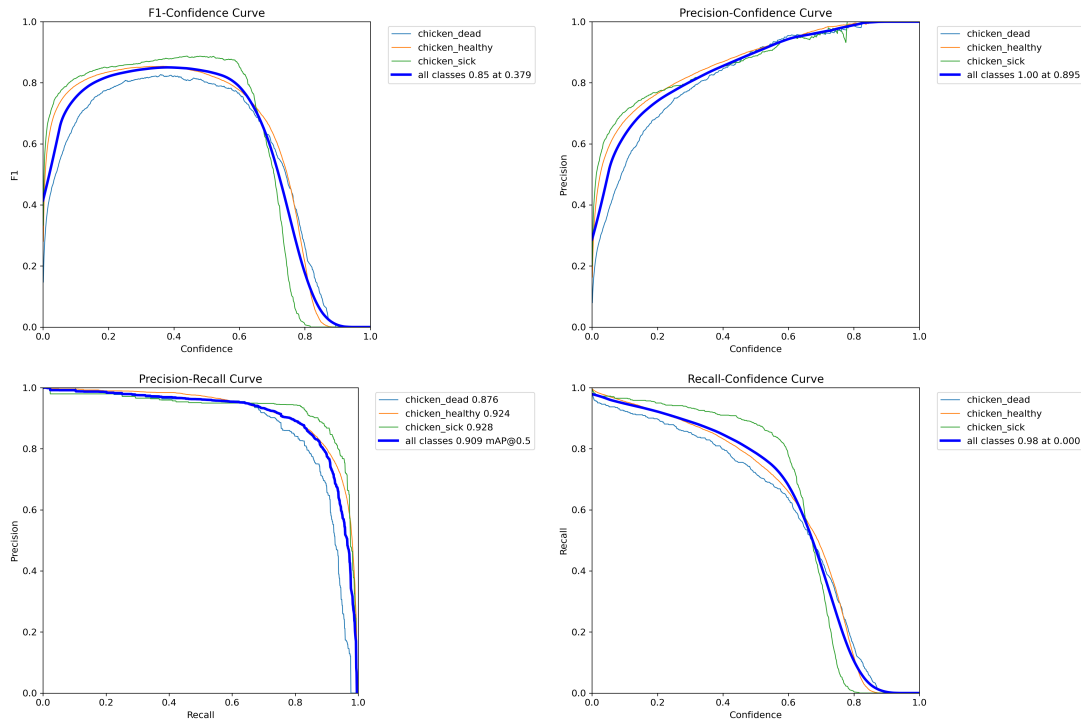


Fig. 5. Mortality Assessment Model Output

multi-stage pipeline improves efficiency, scalability, and real-time performance. Experimental results demonstrate that the system achieves high accuracy across key tasks, particularly fecal disease classification and mortality detection. The integration of edge and cloud computing allows real-time data processing while balancing computational demands. Overall, the proposed framework provides a practical and scalable solution for poultry health monitoring in resource-constrained environments [8], [9].

VIII. FUTURE WORK

Future work will focus on expanding the dataset with real-world farm data collected using the same hardware stack deployed in production. Training on farm-based data annotated by veterinarians is expected to reduce the domain gap and improve model robustness. The Raspberry Pi 4 can also be upgraded to a more advanced AI edge computing platform, such as the NVIDIA Jetson Orin, enabling more stages of the pipeline to run locally without relying on cloud GPU resources. Furthermore, future versions of the fecal disease model may support additional disease categories such as avian flu and malnutrition indicators. The modular design of Poultrix also allows adaptation to other livestock such as goats, cattle, and swine through retraining. Federated learning may also be explored to enable privacy-preserving model improvement across multiple farms.

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